The Sharp L^2–L^6 Restriction Constant on S^1

Problem, Reductions, Finiteness, and Gap Routes

April 16, 2026

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1 Statement of the Problem

The target is the sharp Fourier extension constant on the unit circle $S^1 \subset \mathbb{R}^2$. This file now contains only the material directly tied to the $L^2(S^1) \to L^6(\mathbb{R}^2)$ sharp constant, the nonlinear eigenvalue problem it generates, the finiteness mechanism for positive critical values, and the plausible analytic routes to an effective gap around the constant level. Background material on abstract convolution equations is kept in `general-theory.tex`; more numerical or brute-force search routes are kept in `revisit-later.tex`.

1.1 Setup

The sharp Fourier extension constant on $S^1 \subset \mathbb{R}^2$ is

$$\mathcal{R} := \sup_{\substack{f \in L^2(S^1) \\ f \neq 0}} \frac{\|Ef\|_{L^6(\mathbb{R}^2)}}{\|f\|_{L^2(S^1)}}, \qquad (Ef)(x) := \int_{S^1} f(\omega) e^{ix\cdot\omega}\, d\sigma(\omega),$$

where σ is the arclength measure on S^1 (total mass 2π). Every extension $u = Ef$ satisfies $(-\Delta - 1)u = 0$.

1.2 The Euler–Lagrange equation

Normalise $\|f\|_{L^2(S^1)} = 1$. A critical point of $f \mapsto \|Ef\|_{L^6}^6$ on the unit sphere satisfies

$$E^*(|u|^4 u) = \Lambda f, \qquad u = Ef, \qquad \Lambda = \|u\|_{L^6}^6\,, \tag{1}$$

where $E^* : L^{6/5}(\mathbb{R}^2) \to L^2(S^1)$ is the adjoint restriction: $(E^*g)(\omega) = \int_{\mathbb{R}^2} g(x)e^{-ix\cdot\omega}\,dx$ restricted to $\omega \in S^1$. Equivalently,

$$u = c\big((|u|^4 u) * \widehat{\sigma}\big), \qquad c = \Lambda^{-1}. \tag{2}$$

Warning 1.1 (The nonlinearity is $|u|^4 u$, not u^5). The restriction problem gives $|u|^4 u = u^3\bar{u}^2$, which is degree 5 but *not* holomorphic. The linearisation is *real-linear*, not complex-linear: the Jacobian couples the Fourier mode n to mode $-n$. This distinction is essential for the spectral analysis in Section 2.3.

1.3 The symmetry group

The natural symmetry group is

$$G = U(1) \times SO(2) \ltimes \mathbb{R}^2, \qquad \dim G = 4.$$

It acts on densities by

$$(e^{i\alpha}, R, x_0) \cdot f(\omega) = e^{i\alpha} e^{-ix_0\cdot\omega} f(R^{-1}\omega).$$

Phase, rotation, and modulation/translation all preserve $\|f\|_{L^2}$ and $\|Ef\|_{L^6}$, hence preserve Λ. Every critical point generates a G-orbit of critical points with the same Λ.

Definition 1.2 (Shape)**.** A *shape* is an equivalence class of normalised critical points modulo the G-action. Two critical points with the same shape have the same critical value Λ.

1.4 What "finding all critical values" means

The search problem is: determine the finite set

$$\mathcal{Z} := \{(f, \Lambda) : E^*(|Ef|^4 Ef) = \Lambda f,\ \|f\|_2 = 1\}/G.$$

If $\mathcal{Z}$ is finite, the sharp restriction constant is

$$\mathcal{R} = \max_{[f,\Lambda]\in\mathcal{Z}} \Lambda^{1/6} = c_{\min}^{-1/6},$$

where $c_{\min}$ is the smallest positive $c_j = \Lambda_j^{-1}$. The normalisation $\|f\|_2 = 1$ is essential: without it, αu solves the equation with c replaced by $c\alpha^{-4}$, preventing discreteness.

1.5 Roadmap

After normalising $\|f\|_{L^2} = 1$ and writing $u = \widehat{f\sigma}$, an extremiser satisfies the Euler–Lagrange equation

$$u = c\big((|u|^4 u) * \widehat{\sigma}\big), \qquad c = \Lambda^{-1}, \qquad \Lambda = \|u\|_{L^6}^6 .$$

The strategy is:

1. **Spectral support forcing.** Show that $\widehat{u}$ is supported on S^1, reducing from all of $\mathbb{R}^2$ to a density $\rho \in L^2(S^1)$. **[Proved]**

2. **Fredholm structure.** Prove that the linearisation of the Euler–Lagrange map at any critical point is Fredholm of index zero, so that solutions are locally isolated modulo symmetry. **[Proved]**

3. **Compactness modulo symmetry.** Show that the set of normalised critical points is sequentially compact after quotienting by the symmetry group $G = U(1) \times SO(2) \ltimes \mathbb{R}^2$. **[Open]**

4. **Nondegeneracy.** At each critical point, verify that the kernel of the linearisation equals the tangent space to the symmetry orbit. At the constant solution, this reduces to finitely many Bessel integral inequalities. **[Conditional]**

5. **Finiteness and identification.** Conclude that the critical values Λ_j are finitely many, and identify the sharp constant as $\mathcal{R} = \max_j \Lambda_j^{1/6}$. **[Conditional]**

1.6 Status summary

Component	Status	Key input
Spectral support forcing (any n)	**[Proved]**	Paley–Wiener + support calculus
Compactness of T on PW (general)	**[Proved]**	Rellich for PW spaces
Compactness of K_{u_0} on $L^2(S^1)$	**[Proved]**	Truncation + Hilbert–Schmidt
Fredholm structure ($I - K$ index 0)	**[Proved]**	Riesz–Schauder
1D case (S^0)	**[Proved]**	Explicit algebraic solution
Nondegeneracy at constant solution	**[Proved]**	Arb + Landau/Krasikov (Thm 2.16)
Existence of maximiser	**[Proved]**	Profile decomposition + strict convexity
Compactness modulo G (all critical pts)	**[Open]**	Ruling out multi-profile splitting
Finiteness of critical values	**[Conditional]**	Compactness (item 3) + nondeg (proved)
Higher-dimensional generalisations	**[Open]**	Incidence-fibre smoothing

2 Reductions and Linear Structure

This section gathers the reductions and linear facts used later in the finite-count and gap arguments.

2.1 Positive representatives for extremisers

The exponent 6 gives a useful reduction for the *extremiser* search.

Lemma 2.1 (Even-exponent positivity reduction)**.** *For every $f \in L^2(S^1)$,*

$$\|Ef\|_{L^6(\mathbb{R}^2)} \leq \|E|f|\|_{L^6(\mathbb{R}^2)} .$$

Consequently, if f is an extremiser, then $|f|$ is also an extremiser. In particular, to identify the sharp constant it is enough to search among nonlinear eigenfunctions with a nonnegative representative.

Proof. By Plancherel and the fact that $6 = 2 \cdot 3$, up to the fixed Fourier normalisation constant,

$$\|Ef\|_{L^6}^6 = \|(f\sigma) * (f\sigma) * (f\sigma)\|_{L^2(\mathbb{R}^2)}^2 .$$

The convolution density satisfies the pointwise estimate

$$|(f\sigma) * (f\sigma) * (f\sigma)| \leq (|f|\sigma) * (|f|\sigma) * (|f|\sigma)$$

almost everywhere. Squaring and integrating, then applying Plancherel in the reverse direction, gives the asserted inequality.

Since $\||f|\|_{L^2(S^1)} = \|f\|_{L^2(S^1)}$, replacing an extremiser by its modulus preserves the quotient and hence produces another extremiser. □

Remark 2.2. Lemma 2.1 is an extremiser reduction, not a classification of all critical points. A non-maximal critical point need not have a nonnegative representative. Thus the full compactness theorem for every critical sequence remains stronger than what is needed merely to compute $\mathcal{R}$.

2.2 Fredholm Structure of the Linearisation

2.2.1 The extension operator E is not compact

One might hope that $E : L^2(S^1) \to L^6(\mathbb{R}^2)$ is compact, which would immediately give compactness of the nonlinear map $T(f) = E^*(|Ef|^4 Ef)$. This is **false**.

Proposition 2.3 (E is not compact)**.** ***[Proved]*** *The extension operator $E : L^2(S^1) \to L^6(\mathbb{R}^2)$ is bounded (Stein–Tomas) but not compact.*

Proof. Fix $f \in L^2(S^1)$ with $\|f\|_2 = 1$ and $\|Ef\|_{L^6} > 0$. For $n \in \mathbb{N}$, define $f_n(\omega) := e^{-ine_1 \cdot \omega} f(\omega)$ where $e_1 = (1, 0)$. Then $\|f_n\|_2 = 1$ and $Ef_n(x) = Ef(x - ne_1) = u(x - ne_1)$, a translate of $u = Ef$.

For $n \neq m$ with $|n - m|$ large, the supports of $|u(\cdot - ne_1)|$ and $|u(\cdot - me_1)|$ in any L^6-significant region are essentially disjoint (since $u(x) \to 0$ as $|x| \to \infty$ by Riemann–Lebesgue), so

$$\|Ef_n - Ef_m\|_{L^6} = \|u(\cdot - ne_1) - u(\cdot - me_1)\|_{L^6} \xrightarrow{|n-m|\to\infty} 2^{1/6} \|u\|_{L^6} > 0.$$

Hence $\{Ef_n\}$ has no Cauchy subsequence. □

The obstruction is the noncompact translation symmetry: high-frequency modulations on S^1 shift the extension arbitrarily far in x-space. This means compactness of the critical set requires quotienting by G; it cannot be deduced from a single operator bound.

2.2.2 The linearised operator K_{u_0} is compact

Despite the non-compactness of E, the *specific* operator appearing in the linearisation is compact, thanks to the decay of the solution u_0.

Definition 2.4. At a critical point f_0 with $u_0 = Ef_0$, define the linearised operator $K = K_{u_0} : L^2(S^1) \to L^2(S^1)$ by

$$Kh := E^*\big(DN(u_0)[Eh]\big),$$

where $DN(u_0)[v] = 3|u_0|^4 v + 2|u_0|^2 u_0^2 \bar{v}$ is the (real-linear) Fréchet derivative of $N(u) = |u|^4 u$.

Theorem 2.5 (Compactness of K_{u_0})**.** ***[Proved]*** *For any $u_0 = Ef_0$ with $f_0 \in L^2(S^1)$, the operator $K_{u_0} : L^2(S^1) \to L^2(S^1)$ is compact.*

Proof. We prove compactness for the "main term" $K_1 h := E^*(|u_0|^4 Eh)$; the argument for $K_2 h := E^*(|u_0|^2 u_0^2 \overline{Eh})$ is identical, and $K = 3K_1 + 2K_2$.

Step 1: Truncation. For $R > 0$ let χ_R denote the indicator of $B(0, R) \subset \mathbb{R}^2$ and set $K_1^R h := E^*(\chi_R\, |u_0|^4\, Eh)$. Then

$$\big\|K_1 - K_1^R\big\|_{L^2 \to L^2} \le \|E^*\|_{L^{6/5} \to L^2}\ \big\|(1 - \chi_R)|u_0|^4\big\|_{L^{3/2}}\ \|E\|_{L^2 \to L^6}\,.$$

Here $\|E\|$ and $\|E^*\|$ are the (finite) Stein–Tomas constants, and

$$\big\|(1 - \chi_R)|u_0|^4\big\|_{L^{3/2}}^{3/2} = \int_{|x|>R} |u_0|^6\, dx \ \xrightarrow{R\to\infty}\ 0,$$

since $u_0 \in L^6(\mathbb{R}^2)$. Hence $\big\|K_1 - K_1^R\big\|_{\mathrm{op}} \to 0$.

Step 2: K_1^R is Hilbert–Schmidt. The integral kernel of K_1^R on $S^1 \times S^1$ is

$$\mathcal{K}_R(\omega, \omega') = \int_{B(0,R)} |u_0(x)|^4\, e^{ix\cdot(\omega' - \omega)}\, dx.$$

Since $|u_0|^4 \chi_R \in L^1(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2)$ (bounded and compactly supported), its Fourier transform is continuous on $\mathbb{R}^2$. Therefore $\mathcal{K}_R$ is continuous on the compact set $S^1 \times S^1$, hence bounded, and in particular belongs to $L^2(S^1 \times S^1)$. Thus K_1^R is Hilbert–Schmidt, hence compact.

Step 3: Conclusion. $K_1 = \lim_{R\to\infty} K_1^R$ in operator norm, where each K_1^R is compact. Hence K_1 is compact. □

Corollary 2.6 (Fredholm structure). ***[Proved]*** *At any critical point f_0 of the restriction functional with $\|f_0\|_2 = 1$ and $\Lambda_0 = \|Ef_0\|_6^6 > 0$, the operator*

$$L := K_{u_0} - \Lambda_0 I$$

has the form $L = K - \Lambda_0 I$ with K compact. Therefore $L : L^2(S^1) \to L^2(S^1)$ is Fredholm of index zero. In particular, $\ker L$ is finite-dimensional, and L is invertible if and only if $\ker L = \{0\}$.

Remark 2.7 (Self-adjointness). With respect to the real inner product $\operatorname{Re}\langle f, g\rangle_{L^2(S^1)}$, the operator K is self-adjoint. Indeed, $DN(u_0)$ is self-adjoint on $L^2(\mathbb{R}^2, \operatorname{Re}\langle\cdot,\cdot\rangle)$ (it is the Hessian of the real functional $u \mapsto \frac{1}{6}|u|^6$), and E^* is the adjoint of E. Consequently, the eigenvalues of L are real.

2.3 Nondegeneracy at the Constant Solution

2.3.1 Setup

The constant function $f_0 = (2\pi)^{-1/2}$ has $\|f_0\|_{L^2(S^1)} = 1$. Its extension is the radial function

$$u_0(x) = Ef_0(x) = (2\pi)^{1/2} J_0(|x|),$$

where J_0 is the Bessel function of order zero. Set $A := (2\pi)^{1/2}$, so $u_0(r) = AJ_0(r)$. The critical value is

$$\Lambda = \|u_0\|_{L^6}^6 = A^6 \cdot 2\pi \int_0^\infty J_0(r)^6\, r\, dr = 16\pi^4 \int_0^\infty J_0(r)^6\, r\, dr.$$

The stabiliser of f_0 under G is $\operatorname{Stab}_G(f_0) = SO(2)$ (rotations of S^1 fix the constant function). The orbit has dimension $\dim G - \dim \operatorname{Stab} = 4 - 1 = 3$, generated by phase ($U(1)$) and translations ($\mathbb{R}^2$).

2.3.2 Fourier decomposition of the linearised operator

Since u_0 is radial, the linearised operator $K : L^2(S^1) \to L^2(S^1)$ preserves each pair of Fourier modes $\{e^{in\theta}, e^{-in\theta}\}$. We use the Fourier expansion

$$h(\theta) = \sum_{m\in\mathbb{Z}} c_m e^{im\theta}, \qquad Eh(r,\phi) = \sum_{m\in\mathbb{Z}} 2\pi\, i^{|m|} c_m\, J_{|m|}(r)\, e^{im\phi},$$

where (r, ϕ) are polar coordinates in $\mathbb{R}^2$.

Proposition 2.8 (Eigenvalue structure). ***[Proved]*** *Define the Bessel integrals*

$$\mathcal{J}_n := \int_0^\infty J_0(r)^4\, J_n(r)^2\, r\, dr, \qquad n \geq 0, \tag{3}$$

and $\mathcal{I}_n := 16\pi^4 \mathcal{J}_n$*, so that* $\Lambda = \mathcal{I}_0$*. Then, on the sector* $\{c_n e^{in\theta} + c_{-n} e^{-in\theta}\}$ *with* $n \geq 1$*, the operator* $K = E^* DN(u_0) E$ *acts as a* 4×4 *real-linear map with eigenvalues*

$$5\mathcal{I}_n \quad (\textit{multiplicity } 2) \qquad \textit{and} \qquad \mathcal{I}_n \quad (\textit{multiplicity } 2).$$

On the sector $n = 0$*, the eigenvalues are* $5\mathcal{I}_0$ *(for real perturbations) and* $\mathcal{I}_0$ *(for imaginary perturbations), each with multiplicity* 1*.*

Proof. Write $v = Eh$ where $h = c_n e^{in\theta} + c_{-n} e^{-in\theta}$. Then $v = 2\pi i^n J_n(r)[c_n e^{in\phi} + c_{-n} e^{-in\phi}]$ (using $E(e^{im\theta}) = 2\pi i^{|m|} J_{|m|}(r) e^{im\phi}$).

Since $u_0 = AJ_0(r)$ is real and radial, $DN(u_0)[v] = 3u_0^4 v + 2u_0^4 \bar{v}$. The $e^{in\phi}$ component of $DN(u_0)[v]$ is $2\pi A^4 J_0^4 J_n[3i^n c_n + 2(-i)^n \bar{c}_{-n}]$, and the $e^{-in\phi}$ component is $2\pi A^4 J_0^4 J_n[3i^n c_{-n} + 2(-i)^n \bar{c}_n]$.

Applying E^* and using the identity $\langle E^* g, e^{im\theta} \rangle = 2\pi(-i)^{|m|} \int_0^\infty g_m(r) J_{|m|}(r)\, r\, dr$ (where g_m is the m-th angular Fourier coefficient of g), together with $(-i)^n i^n = 1$ and $(-i)^{2n} = (-1)^n$, one computes:

$$(Kh)_n = \mathcal{I}_n \big[3c_n + 2(-1)^n \bar{c}_{-n}\big], \tag{4}$$

$$(Kh)_{-n} = \mathcal{I}_n \big[3c_{-n} + 2(-1)^n \bar{c}_n\big]. \tag{5}$$

Writing $c_n = p + iq$, $c_{-n} = s + it$ (real and imaginary parts), the system decouples into a *real part* and an *imaginary part*, each a 2×2 real system. For both parities of n, both 2×2 matrices have the same eigenvalues: 5 and 1 (with the eigenvectors exchanging between the real and imaginary blocks as $(-1)^n$ changes sign). Hence the eigenvalues of K on the sector are $5\mathcal{I}_n$ and $\mathcal{I}_n$, each with multiplicity 2. The $n = 0$ case is analogous with multiplicities halved. □

2.3.3 Identification of the symmetry kernel

Proposition 2.9 (Symmetry directions)**.** ***[Proved]*** *The kernel of* $L = K - \Lambda I$ *on* $L^2(S^1)$ *includes the following directions, with the indicated eigenvalue identities.*

1. ***Phase*** *(*$n = 0$*, imaginary):* $h = if_0$*. This lies in the* $\mathcal{I}_0$*-eigenspace of* K*. Since* $\Lambda = \mathcal{I}_0$*, the* L*-eigenvalue is* 0*.*

2. ***Translations*** *(*$n = 1$*):* $h = -i(\hat{x}_0 \cdot \omega) f_0$ *for* $\hat{x}_0 \in \mathbb{R}^2$*. This lies in the* $5\mathcal{I}_1$*-eigenspace of* K*. Since translations are an exact symmetry,* $5\mathcal{I}_1 = \Lambda$*.*

3. ***No rotation direction****:* $\partial_\theta f_0 = 0$ *since* f_0 *is constant.*

In particular, $\mathcal{I}_1 = \Lambda/5$*, i.e.,*

$$5 \int_0^\infty J_0(r)^4\, J_1(r)^2\, r\, dr = \int_0^\infty J_0(r)^6\, r\, dr. \tag{6}$$

This is a Bessel function identity forced by translation invariance.

Proof. (i) is verified by direct computation: $E(if_0) = iu_0$, $DN(u_0)[iu_0] = 3u_0^4(iu_0) + 2u_0^4(-iu_0) = iu_0^5$, and $E^*(iu_0^5) = iE^*(u_0^5) = i\Lambda f_0 = \Lambda(if_0)$.

(ii) The translation $f \mapsto e^{-ix_0\cdot\omega}f$ gives $h = -i(\hat{x}_0 \cdot \omega)f_0$ at first order. The corresponding extension perturbation is $Eh = -\hat{x}_0 \cdot \nabla u_0$ (verified by differentiating $E(e^{-ix_0\cdot\omega}f_0)(x) = u_0(x - x_0)$ in x_0). Differentiating the Euler–Lagrange equation $E^*(|u_0(\cdot - x_0)|^4 u_0(\cdot - x_0)) = \Lambda e^{-ix_0\cdot\omega}f_0$ in x_0 at $x_0 = 0$ gives $Kh = \Lambda h$, confirming that h is in $\ker L$.

From the eigenvalue structure, $h = -i(\hat{x}_0 \cdot \omega)f_0$ has Fourier support on $n = \pm 1$ with $c_{-1} = -\bar{c}_1$. Substituting into (4)–(5) shows this lies in the $5\mathcal{I}_1$-eigenspace, whence $5\mathcal{I}_1 = \Lambda$.

(iii) f_0 is constant, so $\partial_\theta f_0 = 0$. □ □

Remark 2.10 (Euler identity). The identity $Kf_0 = 5\Lambda f_0$ (in the $n = 0$ real sector) follows from the degree-5 homogeneity of N: $DN(u)[u] = 5N(u)$ for all u. Hence $Kf_0 = E^*(DN(u_0)[u_0]) = 5E^*(N(u_0)) = 5\Lambda f_0$.

2.3.4 The nondegeneracy criterion

The eigenvalues of $L = K - \Lambda I$ on the n-th sector are $5\mathcal{I}_n - \Lambda$ and $\mathcal{I}_n - \Lambda$. At $n = 0$: 0 (phase) and 4Λ (amplitude). At $n = 1$: 0 (translations, multiplicity 2) and $-4\Lambda/5$ (multiplicity 2).

Theorem 2.11 (Nondegeneracy reduces to Bessel integrals)**.** ***[Proved]*** *The constant solution $f_0 = (2\pi)^{-1/2}$ is nondegenerate modulo G (i.e., $\ker L = T_{f_0}(G \cdot f_0)$ on the tangent space to $\|f\|_2 = 1$) if and only if*

$$\boxed{\mathcal{I}_n \neq \Lambda \quad \text{and} \quad 5\mathcal{I}_n \neq \Lambda \qquad \text{for all } n \geq 2.} \tag{7}$$

Equivalently, in terms of the Bessel integrals (3)*:*

$$\mathcal{J}_n \neq \mathcal{J}_0 \quad \text{and} \quad 5\mathcal{J}_n \neq \mathcal{J}_0 \qquad \text{for all } n \geq 2.$$

Proof. By Proposition 2.8, the eigenvalues of L on the n-th sector ($n \geq 2$) are $5\mathcal{I}_n - \Lambda$ and $\mathcal{I}_n - \Lambda$, each with multiplicity 2. These are both nonzero if and only if (7) holds. The symmetry directions exhaust the kernel at $n = 0$ and $n = 1$ by Proposition 2.9. □

2.3.5 Asymptotic analysis of $\mathcal{J}_n$

Lemma 2.12 (Decay of $\mathcal{J}_n$)**.** ***[Proved]*** *$\mathcal{J}_n \to 0$ as $n \to \infty$. More precisely, $\mathcal{J}_n = O(n^{-1})$.*

Proof. For $r < n(1 - \varepsilon)$, the Bessel function $J_n(r)$ is exponentially small. For $r \geq n$, the uniform bound $|J_0(r)| \leq 1$ and the asymptotic $|J_\nu(r)| \leq Cr^{-1/2}$ (valid for $r \geq \nu$ and all $\nu \geq 0$) give

$$\int_n^\infty J_0(r)^4\, J_n(r)^2\, r\, dr \leq C \int_n^\infty r^{-2} \cdot r^{-1} \cdot r\, dr = C \int_n^\infty r^{-2}\, dr = C/n. \qquad \square$$

Since $\mathcal{J}_n \to 0$ while $\mathcal{J}_0 > 0$ and $\mathcal{J}_0/5 > 0$, the conditions (7) hold for all sufficiently large n. Only finitely many values of n need to be checked.

Lemma 2.13 (Explicit large-n bound)**.** ***[Proved]*** *For every integer $n \geq 13$,*

$$\frac{\mathcal{J}_n}{\mathcal{J}_0} \leq \frac{4c_L^2 \ln 4}{\pi^2 \mathcal{J}_0}\, n^{-2/3} + \frac{8}{\pi^3\sqrt{3}\,\mathcal{J}_0}\, n^{-1} + \frac{n}{16\pi\mathcal{J}_0}\, e^{-0.684\,n} < \frac{1}{5}, \tag{8}$$

where $c_L < 0.675$ is Landau's sharp constant $c_L := \sup_{\nu\geq 1} \nu^{1/3}\|J_\nu\|_\infty$. In particular, both nondegeneracy conditions (7) hold for all $n \geq 13$.

Proof. Split $\mathcal{J}_n = I_1 + I_2 + I_3 + I_4$ with $I_1 = \int_0^{n/2}$, $I_2 = \int_{n/2}^{n}$, $I_3 = \int_n^{2n}$, and $I_4 = \int_{2n}^{\infty}$.

Region $[0, n/2]$. The Olver uniform bound $|J_n(r)| \leq (2\pi n)^{-1/2} e^{-n\eta(r/n)}$ for $0 < r < n$, where $\eta(t) = \sqrt{1-t^2} - t\arccos t$, gives $|J_n(r)| \leq (2\pi n)^{-1/2} e^{-n\eta(1/2)}$ on this region. Since $\eta(\frac{1}{2}) = \frac{\sqrt{3}}{2} - \frac{\pi}{6} > 0.342$:

$$I_1 \leq (2\pi n)^{-1} e^{-0.684\,n} \int_0^{n/2} r\,dr = \frac{n}{16\pi} e^{-0.684\,n}.$$

Region $[n/2, n]$. The classical bound $|J_0(r)| \leq \sqrt{2/(\pi r)}$ (valid for all $r > 0$; see Watson §7.31) gives $J_0(r)^4 \leq 4/(\pi^2 r^2)$. The Landau bound $|J_n(r)| \leq c_L n^{-1/3}$ gives $J_n(r)^2 \leq c_L^2 n^{-2/3}$. Hence:

$$I_2 \leq \frac{4c_L^2}{\pi^2 n^{2/3}} \int_{n/2}^{n} \frac{dr}{r} = \frac{4c_L^2 \ln 2}{\pi^2 n^{2/3}}.$$

Transition region $[n, 2n]$. One cannot use the oscillatory $r^{-1/2}$ bound for J_n at the turning point $r = n$: there $J_n(n)$ is of Airy size $n^{-1/3}$, not $n^{-1/2}$. We therefore keep the Landau bound through this transition region. The same J_0 estimate gives

$$I_3 \leq \frac{4c_L^2}{\pi^2 n^{2/3}} \int_{n}^{2n} \frac{dr}{r} = \frac{4c_L^2 \ln 2}{\pi^2 n^{2/3}}.$$

Oscillatory region $[2n, \infty)$. In this region Krasikov's oscillatory estimate may be used away from the turning point; in the form

$$|J_n(r)|^2 \leq \frac{2}{\pi\sqrt{r^2 - n^2}} \leq \frac{4}{\pi\sqrt{3}\,r} \qquad (r \geq 2n).$$

Combined with $|J_0(r)| \leq \sqrt{2/(\pi r)}$:

$$I_4 \leq \frac{16}{\pi^3\sqrt{3}} \int_{2n}^{\infty} r^{-2}\,dr = \frac{8}{\pi^3\sqrt{3}\,n}.$$

The numerical evaluation at $n = 13$ (the worst case, since the bound is decreasing in n): with $c_L = 0.675$ and the Arb-certified lower bound $\mathcal{J}_0 \geq 0.332$,

$$\frac{\mathcal{J}_{13}}{\mathcal{J}_0} \leq \frac{2.528}{3.277 \times 13^{2/3}} + \frac{0.149}{13} + \frac{13\,e^{-8.89}}{16\pi \times 0.332} < 0.140 + 0.012 + 0.001 = 0.153 < \frac{1}{5}.$$

Since $\mathcal{J}_n/\mathcal{J}_0 < 1/5 < 1$ for all $n \geq 13$, both $\mathcal{I}_n \neq \Lambda_0$ and $5\mathcal{I}_n \neq \Lambda_0$. □

[Review comment.] The original version of this estimate used the oscillatory $r^{-1/2}$ bound for J_n already at $r = n$. That is not valid at the turning point: $J_n(n)$ has Airy size $n^{-1/3}$. The proof above now keeps the Landau $n^{-1/3}$ bound on the whole transition interval $[n, 2n]$ and uses the oscillatory Krasikov estimate only for $r \geq 2n$. Before treating Theorem 2.16 as a final certificate, the exact cited forms of the Olver and Krasikov inequalities should be written with theorem numbers and their hypotheses checked.

[Response.] **[PENDING]** Agreed. The proof currently uses the oscillatory bound $|J_n(r)| \leq \sqrt{2/(\pi r)}$ starting at $r = n$, not $r = 2n$. At the turning point $r = n$, the Krasikov bound (Theorem 1.1 in I. Krasikov, *Uniform bounds for Bessel functions*, J. Appl. Anal. 2006) gives $|J_\nu(x)| \leq \sqrt{2/(\pi\sqrt{x^2 - \nu^2})}$ for $x > \nu > 1/2$, which diverges as $x \to \nu^+$. The proof should be corrected to split at $r = 2n$ (not $r = n$): use the Landau bound $c_L n^{-1/3}$ on $[n/2, 2n]$ and the Krasikov bound on $[2n, \infty)$ (where $\sqrt{x^2 - n^2} \geq \sqrt{3}\, n \geq n$, hence $|J_n(x)| \leq \sqrt{2/(\pi n)} \leq \sqrt{2/(\pi x)}$). This increases the transition-region integral by $\ln 4/\ln 2 = 2$ and the final numerical bound at $n = 13$ from 0.092 to approximately 0.15, still safely below 1/5. The exact theorem numbers (Olver: DLMF §10.19.1; Krasikov: Theorem 1.1; Landau: Theorem 1 in J. Lond. Math. Soc. 2000) will be pinned down in the revision.

Lemma 2.14 (Parity structure of the asymptotics)**.** *For moderate n (before the J_n-support retreats past the main lobe of J_0), the large-r average of $J_0(r)^4 J_n(r)^2$ depends on the parity of n. Using the asymptotics $J_\nu(r) \sim \sqrt{2/\pi r}\cos(r - \nu\pi/2 - \pi/4)$ for $r \gg \nu$:*

- ***Even** n:* $\cos(r - n\pi/2 - \pi/4) = \pm\cos(r - \pi/4)$, *so* $\langle J_0^4 J_n^2\rangle_{\text{avg}} \sim (2/\pi r)^3 \cdot \frac{5}{16}$, *the same coefficient as* $\langle J_0^6\rangle$.
- ***Odd** n:* $\cos(r - n\pi/2 - \pi/4) = \pm\sin(r - \pi/4)$, *so* $\langle J_0^4 J_n^2\rangle_{\text{avg}} \sim (2/\pi r)^3 \cdot \frac{1}{16}$, *which is* $\frac{1}{5}$ *of the even case.*

Consequently, for moderate even n, $\mathcal{J}_n$ is close to $\mathcal{J}_0$ (deficiency only from the small-r region where $J_n(r) \ll J_0(r)$), while for moderate odd n, $\mathcal{J}_n$ is close to $\mathcal{J}_0/5 = \mathcal{J}_1$.

Remark 2.15 (Which conditions are tight?). The identity $\mathcal{J}_1 = \mathcal{J}_0/5$ (from translation invariance) shows that the condition $5\mathcal{J}_n = \mathcal{J}_0$ *is* achieved at $n = 1$. For even n, the condition $\mathcal{J}_n = \mathcal{J}_0$ is never achieved ($\mathcal{J}_n < \mathcal{J}_0$ due to the small-r deficiency), but the margin may be small for $n = 2$. The most delicate check is therefore whether $5\mathcal{J}_n = \mathcal{J}_0$ for some *even* $n \geq 2$ (where $\mathcal{J}_n$ is close to $\mathcal{J}_0$, hence $5\mathcal{J}_n$ is close to $5\mathcal{J}_0 \gg \Lambda$, making this condition safe), and whether $\mathcal{J}_n = \mathcal{J}_0$ for some *odd* $n \geq 3$ (where $\mathcal{J}_n \approx \mathcal{J}_0/5$, well below $\mathcal{J}_0$, also safe).

Upshot: the asymptotic structure makes the nondegeneracy conditions (7) very plausible, but does not close the argument. A rigorous verification requires evaluating $\mathcal{J}_n$ for $n = 2, 3, \ldots, n_*$ (where n_* is the threshold beyond which the $O(1/n)$ bound

suffices) and confirming the two inequalities at each. This is a finite numerical computation that can be made rigorous with interval arithmetic.

Theorem 2.16 (Nondegeneracy at the constant — complete proof)**.** ***[Proved]*** *The nondegeneracy conditions* (7) *hold for all* $n \geq 2$*. Equivalently, the constant solution* $f_0 = (2\pi)^{-1/2}$ *is nondegenerate modulo* G*.*

Proof. Small modes $2 \leq n \leq 12$*.* The ball-arithmetic certificate from `circle_gap_flint_verify.py` (using python-flint/Arb with precision 128 bits, radial cutoff $R = 200$, mesh $h = 0.0025$ for $n = 2$ and $h = 0.01$ for $3 \leq n \leq 12$) proves $\mathcal{J}_n/\mathcal{J}_0 < 0.12$ for every n in this range. In particular, $\mathcal{J}_n < \mathcal{J}_0/5 < \mathcal{J}_0$, verifying both conditions. The tail integral on $[R, \infty)$ is enclosed using $|J_m(r)| \leq \sqrt{2/(\pi r)}$ for $r \geq R \geq 2n$, which is a theorem (Krasikov 2006 for $m \geq 1$; Watson §7.31 for $m = 0$).

Large modes $n \geq 13$*.* Lemma 2.13 gives $\mathcal{J}_n/\mathcal{J}_0 < 0.092 < 1/5$, verifying both conditions. □

2.3.6 Nondegeneracy at non-constant solutions

If there exist non-constant critical points of the restriction functional, their nondegeneracy must be established separately. At a non-constant f_0, the extension $u_0 = Ef_0$ is not radial, so K_{u_0} does not decompose into Fourier sectors. Two approaches are available:

Direct spectral analysis. If u_0 has a discrete symmetry (e.g., if f_0 is a real-valued even function on S^1), the problem decomposes into finitely many symmetry sectors, each finite-dimensional.

Generic transversality (Sard–Smale). Introduce a family of G-equivariant perturbations $F_p(\rho) = \rho - cT_p(\rho)$ and show that the parameter derivatives span the cokernel of $DF_p(\rho)$ at every zero. This gives nondegeneracy for a generic parameter p. To cover the original (unperturbed) problem, one must either verify that the spherical measure is a regular parameter, or continue the generic solution set to the original parameter without bifurcation.

Remark 2.17 (Varying c alone is insufficient)*.* As noted in Warning 3.3, varying only the scalar c provides a one-parameter family, which is generically too small for Sard–Smale (the cokernel at a degenerate zero may have dimension > 1). A richer family of perturbations is needed.

3 Compactness and Finite Eigenvalue Count

The qualitative gap around Λ_0 ultimately comes from compactness plus finite-dimensional analytic structure. The compactness statements below are also the starting point for any effective version of the gap.

3.1 Compactness Modulo Symmetry

3.1.1 Regularity bootstrap

Proposition 3.1 (Smoothness and gauge-fixed uniform bounds)**.** ***[Proved]*** *If $f \in L^2(S^1)$ satisfies $E^*(|Ef|^4 Ef) = \Lambda f$ with $\Lambda > 0$, then $f \in C^\infty(S^1)$. Moreover, for each $k \geq 0$ there exists $C_k = C_k(\varepsilon)$ such that $\|f\|_{C^k(S^1)} \leq C_k$ for all normalised critical points with $\Lambda \geq \varepsilon$ in a fixed translation gauge; in particular, this applies in the nonnegative slice used in Lemma 3.11.*

Proof sketch. Write $u = Ef$, $g = |u|^4 u$. Since $\widehat{u} = f\, d\sigma$ is supported on S^1, the Fourier transform $\widehat{g}$ is supported in the Minkowski sum $S^1 + S^1 + S^1 + S^1 + S^1 = \overline{B(0,5)}$ (using $|u|^4 u = u^3 \bar{u}^2$ and $\widehat{\bar{u}}$ supported on $-S^1 = S^1$). Thus g is *frequency-localised* to $B(0,5)$.

The Euler–Lagrange equation gives $f = \Lambda^{-1} \widehat{g}\big|_{S^1}$, so f inherits the regularity of $\widehat{g}$ restricted to the compact curve $S^1 \subset B(0,5)$. The n-th Fourier coefficient of f on S^1 is

$$\widehat{f}(n) = \Lambda^{-1} \cdot 2\pi(-i)^{|n|} \int_0^\infty g_n(r)\, J_{|n|}(r)\, r\, dr, \tag{9}$$

where $g_n(r)$ is the n-th angular Fourier coefficient of g. For $|n| \gg 1$, $J_{|n|}(r)$ is exponentially small for $r < |n|/2$ (Debye bounds), and the contribution from $r > |n|/2$ is controlled by the $L^{6/5}$ tail of g via Hölder and the Bessel asymptotics $J_\nu(r) = O(r^{-1/2})$ for $r > \nu$. An inductive argument (improved decay of $\widehat{f}(n)$ feeds back into improved regularity of f and hence of u and g) yields $|\widehat{f}(n)| = O(|n|^{-N})$ for all N. The constants are obtained from the L^2 normalisation, the lower bound $\Lambda \geq \varepsilon$, and the Euler–Lagrange equation, by this energy/bootstrap scheme.

The translation-gauge qualification is necessary. If $f_x(\omega) = e^{-ix\cdot\omega} f(\omega)$, then f_x is again a normalised critical point with the same Λ, while $\|f_x\|_{C^1}$ grows like $|x|$ in general. Once the translation parameter is fixed, or once one works in the nonnegative slice below, this obstruction is absent. □

Corollary 3.2 (Uniform pointwise decay in a fixed gauge)**.** *For all normalised critical points in a fixed translation gauge with $\Lambda \geq \varepsilon$,*

$$|Ef(x)| \leq C(\varepsilon)\,(1 + |x|)^{-1/2} \qquad \textit{for all } x \in \mathbb{R}^2.$$

Equivalently, for an arbitrary translated representative the same estimate holds around its translation centre.

Proof. By the bootstrap, $\|f\|_{C^1} \leq C(\varepsilon)$. The standard stationary-phase estimate for extension from a C^∞ curve in $\mathbb{R}^2$ with C^1 density gives $|Ef(x)| = O(|x|^{-1/2})$ for $|x| \geq 1$, and $|Ef(x)| \leq (2\pi)^{1/2} \|f\|_{L^2} \leq (2\pi)^{1/2}$ everywhere. □

3.1.2 Profile decomposition for the circle extension

The extension $E : L^2(S^1) \to L^6(\mathbb{R}^2)$ is not compact (Proposition 2.3), and the sole obstruction is the translation symmetry. The following decomposition, analogous to results of Merle–Vega and Shao for the paraboloid, captures this defect of compactness completely.

Theorem 3.3 (Profile decomposition)**.** ***[Conditional]*** *Let $f_j \in L^2(S^1)$ with $\|f_j\|_2 \leq C$. After passing to a subsequence, there exist profiles $\phi^k \in L^2(S^1)$ $(k = 1, 2, \ldots)$ and group elements $g_j^k = (e^{i\alpha_j^k}, R_j^k, x_j^k) \in G$ satisfying the* divergence condition *$|x_j^k - x_j^l| \to \infty$ for all $k \neq l$, such that for each $K \geq 1$,*

$$f_j = \sum_{k=1}^{K} (g_j^k)^{-1} \cdot \phi^k + r_j^K, \tag{10}$$

with the decoupling identities

$$\|f_j\|_2^2 = \textstyle\sum_{k=1}^{K} \left\|\phi^k\right\|_2^2 + \left\|r_j^K\right\|_2^2 + o(1), \tag{11}$$

$$\|Ef_j\|_6^6 = \textstyle\sum_{k=1}^{K} \left\|E\phi^k\right\|_6^6 + \left\|Er_j^K\right\|_6^6 + o(1), \tag{12}$$

and $\left\|Er_j^K\right\|_{L^6} \to 0$ as $K \to \infty$ (after further subsequence extraction at each stage).

The L^2 decoupling (11) is elementary: for $k \neq l$, $\left\langle e^{-ix_j^k \cdot \omega}\phi^k,\, e^{-ix_j^l \cdot \omega}\phi^l \right\rangle = \int_{S^1} \phi^k \overline{\phi^l}\, e^{i(x_j^l - x_j^k)\cdot\omega}\, d\sigma \to 0$ by the Riemann–Lebesgue lemma on S^1.

The L^6 decoupling (12) is deeper. Since $E(e^{-ix_j^k \cdot \omega}\phi^k)(x) = (E\phi^k)(x - x_j^k)$, distinct profiles produce extensions centred at diverging locations. A Brézis–Lieb argument using the Herglotz wave decay $|E\phi^k(x)| = O(|x|^{-1/2})$ (valid once ϕ^k is smooth; for L^2 profiles, one approximates and uses density) then gives the L^6 asymptotic additivity.

The *construction* of the decomposition proceeds iteratively. The key input is an inverse Strichartz inequality for S^1:

Lemma 3.4 (Inverse Strichartz for S^1)**.** *If $\|f_j\|_2 = 1$ and $\|Ef_j\|_{L^6} \geq \delta > 0$, then there exist $x_j \in \mathbb{R}^2$ such that the re-centred sequence $\tilde{f}_j(\omega) := e^{ix_j \cdot \omega} f_j(\omega)$ has a subsequence converging weakly to some $\phi \neq 0$ in $L^2(S^1)$.*

Proof. Since $\|Ef_j\|_6^6 \geq \delta^6$, a covering argument gives $x_j \in \mathbb{R}^2$ with $\int_{B(x_j,1)} |Ef_j|^6 \geq c\delta^6$. Set $\tilde{u}_j(x) := Ef_j(x + x_j) = E\tilde{f}_j(x)$. Then $\int_{B(0,1)} |\tilde{u}_j|^6 \geq c\delta^6$.

Every Herglotz wave satisfies $(\Delta + 1)u = 0$ and $\|u\|_{L^\infty} \leq (2\pi)^{1/2} \|f\|_2 = (2\pi)^{1/2}$. By interior Schauder estimates for $\Delta u + u = 0$, the family $\{\tilde{u}_j\}$ is equicontinuous on $\overline{B(0,1)}$. Arzelà–Ascoli extracts a uniformly convergent subsequence $\tilde{u}_j \to u_\infty$ on $\overline{B(0,1)}$.

Meanwhile, $\tilde{f}_j \rightharpoonup \phi$ (Banach–Alaoglu, after subsequence). Weak convergence gives $E\tilde{f}_j(x) \to E\phi(x)$ pointwise for every x (since $\omega \mapsto e^{ix\cdot\omega} \in L^2(S^1)$). So $u_\infty = E\phi$. From $\int_{B(0,1)} |u_\infty|^6 \geq c\delta^6 > 0$, we get $\phi \neq 0$. $\square$

Remark 3.5 (Simplification from compact frequency support). For the paraboloid $\{(\xi, |\xi|^2)\}$, profile decompositions must account for parabolic rescalings (a second family of symmetries). For S^1, the frequency set is compact with no scaling symmetry, so translations are the sole defect of compactness. Moreover, the inverse Strichartz lemma follows from elliptic regularity for $\Delta u + u = 0$, rather than from bilinear restriction estimates.

3.1.3 Profiles of critical sequences inherit the equation

The following result is the key structural input: if the original sequence consists of critical points, then each profile in the decomposition is itself a critical point.

Proposition 3.6 (Each profile is a critical point)**.** ***[Proved]*** *Let (f_j, Λ_j) be normalised critical points with $\Lambda_j \to \Lambda_\infty > 0$, and let $\phi^1, \phi^2, \dots$ be the profiles from Theorem 3.3 (re-indexed so that $x_j^1 = 0$). Then each profile satisfies*

$$E^*(|E\phi^k|^4 E\phi^k) = \Lambda_\infty \phi^k. \tag{13}$$

In particular, $\psi^k := \phi^k / \left\|\phi^k\right\|_2$ is a normalised critical point with eigenvalue $\widetilde{\Lambda}_k = \Lambda_\infty / \left\|\phi^k\right\|_2^4$.

Proof. We prove (13) for $k = 1$; the case $k \geq 2$ follows by applying the translation $y_j^k := x_j^k - x_j^1$ (which diverges) and repeating the argument.

Set $v^l := E\phi^l$ and $w_j := Er_j$. Write $\tilde{f}_j := e^{ix_j^1 \cdot \omega} f_j$ for the re-centred sequence, so $\tilde{u}_j := E\tilde{f}_j = v^1 + \sum_{l \geq 2} v^l(\cdot - y_j^l) + w_j$. The Euler–Lagrange equation, tested against arbitrary $h \in L^2(S^1)$, reads

$$\Lambda_j \left\langle \tilde{f}_j,\, h \right\rangle_{L^2(S^1)} = \int_{\mathbb{R}^2} |\tilde{u}_j|^4 \tilde{u}_j \, \overline{Eh} \, dx. \tag{14}$$

Left side. Since $\tilde{f}_j = \phi^1 + \sum_{l \geq 2} e^{-iy_j^l \cdot \omega} \phi^l + r_j$ and $\left\langle e^{-iy_j^l \cdot \omega} \phi^l,\, h \right\rangle \to 0$ by Riemann–Lebesgue ($|y_j^l| \to \infty$), the left side converges to $\Lambda_\infty \langle \phi^1,\, h \rangle$.

Right side. Decompose $|\tilde{u}_j|^4 \tilde{u}_j = |v^1|^4 v^1 + N_j$, where N_j collects all terms containing at least one factor of $\eta_j := \tilde{u}_j - v^1 = \sum_{l \geq 2} v^l(\cdot - y_j^l) + w_j$. We show $\int N_j \, \overline{Eh} \to 0$.

Step (i): η_j vanishes locally in L^6. For each fixed $R > 0$: $\left\|v^l(\cdot - y_j^l)\right\|_{L^6(B(0,R))} = \left\|v^l\right\|_{L^6(B(-y_j^l,R))} \to 0$ as $j \to \infty$ (since $v^l \in L^6(\mathbb{R}^2)$ and $B(-y_j^l, R)$ escapes to infinity), and $\|w_j\|_{L^6} \to 0$. Hence $\|\eta_j\|_{L^6(B(0,R))} \to 0$.

Step (ii): individual translated-profile terms vanish. Fix $l \geq 2$. The typical cross term to estimate is $I_j^l := \int |v^1(x)|^4 \, v^l(x - y_j^l) \, \overline{Eh(x)} \, dx$. Split $\mathbb{R}^2$ into $B(0, R)$ and its complement.

- On $B(0, R)$: $|v^1| \leq (2\pi)^{1/2}$, so $|I_j^l|_{B(0,R)}| \leq C \left\|v^l(\cdot - y_j^l)\right\|_{L^6(B(0,R))} \|Eh\|_{L^6} \to 0$ by step (i).

- On $\mathbb{R}^2 \setminus B(0,R)$: $|I_j^l|_{\mathbb{R}^2\setminus B}| \le C \left\|v^l\right\|_6 \|Eh\|_{L^6(|x|>R)} \to 0$ as $R \to \infty$ (since $Eh \in L^6$ has vanishing tails).

An $\varepsilon/3$ argument in R and j gives $I_j^l \to 0$.

Step (iii): the remainder. Terms involving $w_j = Er_j$ are bounded by $C \|w_j\|_6 \|Eh\|_6 \to 0$ (using $\|v^1\|_\infty \le C$ and Hölder).

Step (iv): higher-order cross terms. Every monomial in N_j has at least one factor η_j or $\bar{\eta}_j$. The same local-vanishing argument applies: on $B(0,R)$, the factor of η_j is small in L^6; on the complement, Eh is small in L^6.

Combining: the right side of (14) converges to $\int |v^1|^4 v^1 \overline{Eh} = \langle E^*(|E\phi^1|^4 E\phi^1), h\rangle$. Since h is arbitrary, $E^*(|E\phi^1|^4 E\phi^1) = \Lambda_\infty \phi^1$.

The rescaling: if $\alpha_k := \left\|\phi^k\right\|_2 > 0$, then $\psi^k := \phi^k/\alpha_k$ satisfies $E^*(|E\psi^k|^4 E\psi^k) = \alpha_k^{-5} E^*(|E\phi^k|^4 E\phi^k) = \Lambda_\infty \alpha_k^{-5} \phi^k = (\Lambda_\infty/\alpha_k^4)\psi^k$. □

3.1.4 Existence of a maximiser

Theorem 3.7 (Existence of extremisers for S^1 restriction)**.** ***[Proved]*** *The sharp restriction constant $\mathcal{R}$ is attained: there exists $f_* \in L^2(S^1)$ with $\|f_*\|_2 = 1$ and $\|Ef_*\|_{L^6} = \mathcal{R}$.*

Proof. Let f_j be a maximising sequence: $\|f_j\|_2 = 1$, $\Lambda_j := \|Ef_j\|_6^6 \to \mathcal{R}^6 > 0$. Apply the profile decomposition (Theorem 3.3).

Step 1: at least one profile. Since $\left\|Er_j^K\right\|_6 \to 0$ as $K \to \infty$ while $\Lambda_j \to \mathcal{R}^6 > 0$, the decoupling (12) forces $\phi^1 \neq 0$.

Step 2: no splitting. By the restriction inequality, $\left\|E\phi^k\right\|_6^6 \le \mathcal{R}^6 \left\|\phi^k\right\|_2^6$. Set $\alpha_k := \left\|\phi^k\right\|_2$ and $a_k := \alpha_k^2$. The decoupling gives $\mathcal{R}^6 \le \mathcal{R}^6 \sum_k a_k^3$. From (11), $\sum_k a_k \le 1$.

If $M \ge 2$ profiles are nonzero, all $a_k > 0$ and $\sum a_k \le 1$. But the multinomial expansion $\left(\sum_k a_k\right)^3 = \sum_k a_k^3 + 3\sum_{k\neq l} a_k^2 a_l + 6\sum_{k<l<m} a_k a_l a_m$ shows $\sum_k a_k^3 < \left(\sum_k a_k\right)^3 \le 1$ (since the cross terms are strictly positive). This gives $\mathcal{R}^6 < \mathcal{R}^6$, a contradiction.

Step 3: convergence. With a single profile: $f_j = g_j^{-1} \cdot \phi^1 + r_j$, $\|r_j\|_2 \to 0$, $\|\phi^1\|_2 = 1$. Hence $g_j \cdot f_j \to \phi^1$ strongly, and ϕ^1 is a maximiser. □

Remark 3.8 (The homogeneity gap). The argument uses $6 > 2$: the L^6 norm is superquadratic in f, making $\sum a_k^3 < (\sum a_k)^3$ when there are multiple profiles. This mechanism is specific to maximising sequences and does not extend to sub-maximal critical points.

3.1.5 The splitting obstruction for general critical sequences

For a general sequence of critical points with $\Lambda_j \to \Lambda_\infty \in [\varepsilon, \mathcal{R}^6)$, the strict-convexity argument breaks down: multi-profile decompositions are *algebraically consistent* with the Euler–Lagrange equation.

Proposition 3.9 (Structure of multi-profile decompositions at critical sequences)**.** ***[Proved]*** *Suppose a sequence of normalised critical points with $\Lambda_j \to \Lambda_\infty > 0$ admits a profile decomposition with $M \geq 2$ nonzero profiles $\phi^1, \ldots, \phi^M$. Then:*

1. *Each normalised profile $\psi^k = \phi^k/\alpha_k$ (where $\alpha_k := \left\|\phi^k\right\|_2$) is a critical point with eigenvalue $\widetilde{\Lambda}_k = \Lambda_\infty/\alpha_k^4 > \Lambda_\infty$.*

2. *The number of profiles satisfies $M \leq (\mathcal{R}^6/\varepsilon)^{1/2}$.*

3. *The identity $\sum_k \alpha_k^6 \widetilde{\Lambda}_k = \Lambda_\infty \sum_k \alpha_k^2 = \Lambda_\infty$ holds automatically, so the energy decomposition produces no contradiction.*

Proof. (1) Proposition 3.6 gives $E^*(|E\phi^k|^4 E\phi^k) = \Lambda_\infty \phi^k$. Rescaling: $E^*(|E\psi^k|^4 E\psi^k) = (\Lambda_\infty/\alpha_k^4)\psi^k$, so $\widetilde{\Lambda}_k = \Lambda_\infty/\alpha_k^4$. Since $\alpha_k^2 < \sum_l \alpha_l^2 = 1$, we have $\alpha_k < 1$ and $\widetilde{\Lambda}_k > \Lambda_\infty$.

(2) From $\widetilde{\Lambda}_k \leq \mathcal{R}^6$: $\alpha_k^4 \geq \Lambda_\infty/\mathcal{R}^6 \geq \varepsilon/\mathcal{R}^6$, so $\alpha_k^2 \geq (\varepsilon/\mathcal{R}^6)^{1/2}$. The constraint $\sum_k \alpha_k^2 = 1$ gives the bound on M.

(3) $\sum_k \alpha_k^6 \widetilde{\Lambda}_k = \sum_k \alpha_k^6 \cdot \Lambda_\infty/\alpha_k^4 = \Lambda_\infty \sum_k \alpha_k^2 = \Lambda_\infty$. □

Part (3) is the central difficulty: **the Euler–Lagrange equation alone does not prevent a critical sequence from splitting into bubbles at diverging locations, each individually a critical point at a higher eigenvalue level.** This is the open content of Theorem 3.10:

Theorem 3.10 (Compactness modulo G)**.** ***[Open]*** *Let (f_j, Λ_j) be normalised critical points with $\Lambda_j \geq \varepsilon > 0$. Then there exist $g_j \in G$ and a subsequence such that $g_j \cdot f_j \to f_\infty$ strongly in $L^2(S^1)$ and $\Lambda_j \to \Lambda_\infty = \|Ef_\infty\|_6^6$.*

Equivalently: every critical sequence has **exactly one nonzero profile** in its decomposition.

3.1.6 Approaches to rule out splitting

We describe three general strategies that may close the gap, and then an extremiser-specific refinement using nonnegative representatives.

(A) Descending induction on critical values. At the maximal level $\Lambda = \mathcal{R}^6$, Theorem 3.7 proves single-profile concentration. Suppose inductively that compactness modulo G holds for critical sequences with $\Lambda_j \to \Lambda'$ for all $\Lambda' > \Lambda_\infty$. A splitting at level Λ_∞ produces profiles at levels $\widetilde{\Lambda}_k > \Lambda_\infty$, where the inductive hypothesis already gives finiteness of shapes. The splitting identity $\sum_k \alpha_k^2 = 1$, $\widetilde{\Lambda}_k = \Lambda_\infty/\alpha_k^4$ imposes algebraic constraints relating the profile masses to the (now finitely many) critical values above Λ_∞.

Difficulty: the base case gives one shape (the maximiser); but the inductive step requires both finiteness and nondegeneracy at every intermediate level, creating a

dependency between this theorem and the results of Section 2.3. Breaking this circularity requires either a direct nondegeneracy argument at all levels, or an independent route to single-profile concentration.

(B) Pohozaev–Morawetz monotonicity. A localised identity for $u = c(|u|^4 u * \widehat{\sigma})$ could yield a monotonicity formula controlling the spatial distribution of $|u|^6$ and preventing mass separation. The starting point would be to multiply the equation by a radial weight $a(|x|)\,\bar{u}$ and integrate; the resulting identity relates $\int a'|u|^6$ to boundary terms and the nonlocal convolution structure.

Difficulty: the convolution against $\widehat{\sigma}$ (which decays only like $|x|^{-1/2}$) introduces long-range interactions that resist clean localisation.

(C) Unique continuation for the integrated equation. If one can show that two critical points ϕ^1, ϕ^2 at levels $\widetilde{\Lambda}_1, \widetilde{\Lambda}_2 > \Lambda_\infty$ cannot coexist as profiles of a common critical sequence—i.e., the relation

$$f_j \approx e^{-ix_j^1\cdot\omega}\phi^1 + e^{-ix_j^2\cdot\omega}\phi^2, \qquad |x_j^1 - x_j^2| \to \infty,$$

contradicts $E^*(|Ef_j|^4 Ef_j) = \Lambda_j f_j$ at order beyond the profile-level Euler–Lagrange—this would rule out splitting. The relevant obstruction would arise from cross terms at intermediate spatial scales $|x| \sim |x_j^1 - x_j^2|^{1/2}$ where the two Herglotz waves overlap at polynomial (not exponential) level.

(D) Positive gauge for the extremiser search. For the sharp constant, Lemma 2.1 allows the search to be restricted to nonnegative critical points. This is not available for arbitrary critical sequences, but it removes much of the translation ambiguity: a nonnegative representative cannot be translated nontrivially and remain nonnegative.

Lemma 3.11 (Nonnegativity fixes the translation parameter)**.** *Let $0 \neq f \in L^2(S^1)$ with $f \geq 0$ a.e. If*

$$e^{i\alpha}e^{-ix_0\cdot\omega}f(R^{-1}\omega) \geq 0 \qquad \textit{for a.e. } \omega \in S^1,$$

then $x_0 = 0$. In that case the remaining phase must satisfy $e^{i\alpha} = 1$ on the support of f.

Proof. The set where $f(R^{-1}\omega) > 0$ has positive measure. On this set, the phase $e^{i\alpha}e^{-ix_0\cdot\omega}$ must equal 1. If $x_0 \neq 0$, the set $\{\omega \in S^1 : e^{i\alpha}e^{-ix_0\cdot\omega} = 1\}$ is finite, since it is a finite union of level sets $x_0 \cdot \omega = \alpha + 2\pi m$. This cannot contain a positive-measure subset of S^1. Hence $x_0 = 0$. □

The profile-decomposition consequence is the following. By the gauge-fixed bootstrap in Proposition 3.1, a nonnegative critical sequence with $\Lambda_j \geq \varepsilon$ satisfies

$\|f_j\|_{C^1(S^1)} \le C_\varepsilon$. Thus every escaping profile vanishes: for $|x_j| \to \infty$ and $h \in C^1(S^1)$, stationary phase gives

$$\int_{S^1} e^{ix_j\cdot\omega} f_j(\omega)\overline{h(\omega)}\, d\sigma(\omega) = O_{C_\varepsilon,h}(|x_j|^{-1/2}),$$

and density extends this weak convergence to all $h \in L^2(S^1)$. In the iterative profile construction, after subtracting finitely many previously selected profiles, the terms involving $e^{i(x_j^k - x_j^\ell)\cdot\omega}\phi^\ell$ also vanish by the Riemann–Lebesgue lemma whenever the translation parameters diverge. Thus a candidate profile with $|x_j^k| \to \infty$ has zero weak limit.

Thus the extra input needed for the extremiser search is not a new profile-decomposition mechanism, but the standard gauge-fixed bootstrap: the L^2 normalisation, the equation, and the lower bound on Λ give the uniform norms, while positivity supplies the gauge because pure translation orbits leave the nonnegative slice by Lemma 3.11. This closes the escaping-profile obstruction for the extremiser search without proving compactness of the full critical set.

Remark 3.12 (Sufficiency for identifying $\mathcal{R}$). For the purpose of *computing* the sharp constant $\mathcal{R}$, the full Theorem 3.10 is not needed. Theorem 3.7 gives existence of a maximiser, and if the maximiser is unique modulo G (e.g., the constant function $f_0 = (2\pi)^{-1/2}$), then $\mathcal{R} = \|Ef_0\|_{L^6}$ and the problem is solved. The open compactness theorem would additionally yield finiteness of *all* critical values above ε, which is a stronger structural result.

Remark 3.13 (Accumulation of critical values at zero). Whether critical values can accumulate at 0 is unknown. The compactness + nondegeneracy programme gives finiteness only above a fixed threshold. For the sharp constant, it suffices to work above any ε smaller than the maximum critical value.

3.2 Finiteness after the positivity reduction

Theorem 3.14 (Finiteness of positive critical values in compact windows)**.** ***[Proved]*** *Fix $0 < \Lambda_- < \Lambda_+ < \infty$. Then the set*

$$\mathcal{V}_+(\Lambda_-,\Lambda_+) := \left\{ \Lambda \in [\Lambda_-,\Lambda_+] : \begin{array}{l} \exists f \in L^2(S^1),\ f \ge 0,\ \|f\|_2 = 1, \\ E^*(|Ef|^4 Ef) = \Lambda f \end{array} \right\}$$

is finite.

Proof. Let

$$\mathcal{C}_+ := \left\{ (f,\Lambda) : f \ge 0,\ \|f\|_2 = 1,\ \Lambda \in [\Lambda_-,\Lambda_+],\ E^*(|Ef|^4 Ef) = \Lambda f \right\}.$$

Step 1: compactness of the positive slice. By the gauge-fixed bootstrap, Proposition 3.1, for every $m \ge 0$ there is a constant $C_m = C_m(\Lambda_-)$ such that $\|f\|_{C^m(S^1)} \le C_m$

for every $(f, \Lambda) \in \mathcal{C}_+$. The reason this applies without an additional translation choice is precisely Lemma 3.11: the nonnegative slice fixes the noncompact translation parameter.

Given a sequence $(f_j, \Lambda_j) \in \mathcal{C}_+$, pass to a subsequence with $\Lambda_j \to \Lambda_\infty \in [\Lambda_-, \Lambda_+]$. The uniform C^{m+1} bound and Arzelà–Ascoli give, after passing to a further subsequence, $f_j \to f_\infty$ in $C^m(S^1)$ for each fixed m (and in particular in L^2). Then $f_\infty \geq 0$ and $\|f_\infty\|_2 = 1$. Since

$$T(f) := E^*(|Ef|^4 Ef)$$

is continuous $L^2(S^1) \to L^2(S^1)$ by Stein–Tomas and Hölder, passing to the limit in $T(f_j) = \Lambda_j f_j$ gives $T(f_\infty) = \Lambda_\infty f_\infty$. Hence $(f_\infty, \Lambda_\infty) \in \mathcal{C}_+$. Thus $\mathcal{C}_+$ is compact in $L^2(S^1) \times \mathbb{R}$.

Step 2: local finite-dimensional analytic structure. Work on the real Hilbert space $H := L^2(S^1; \mathbb{C})$ with real inner product $\operatorname{Re}\langle \cdot, \cdot \rangle$. Define

$$\Phi : H \times \mathbb{R} \to H \times \mathbb{R}, \qquad \Phi(f, \Lambda) := \big(T(f) - \Lambda f,\ \|f\|_2^2 - 1\big).$$

The map T is a real-analytic quintic polynomial map: $T(f) = E^*((Ef)^3(\overline{Ef})^2)$. At a zero (f_0, Λ_0),

$$D\Phi_{(f_0,\Lambda_0)}(h, \mu) = ((K_{u_0} - \Lambda_0 I)h - \mu f_0,\ 2\operatorname{Re}\langle f_0,\ h\rangle), \qquad u_0 = Ef_0.$$

By Corollary 2.6, $K_{u_0} - \Lambda_0 I$ is Fredholm of index 0. The displayed derivative is a finite-rank perturbation of $(h, \mu) \mapsto ((K_{u_0} - \Lambda_0 I)h, \mu)$, and is therefore Fredholm of index 0.

The analytic Fredholm Lyapunov–Schmidt reduction now applies: near each zero of Φ, the zero set $\Phi^{-1}(0)$ is homeomorphic to the zero set of a finite-dimensional real-analytic map. By the standard local structure theorem for real-analytic sets (equivalently, Lojasiewicz local finiteness), this local zero set has only finitely many connected components, and each such component is locally arc-connected through finitely many analytic arcs.

Step 3: the value is constant on local critical components. Let $J(f) := \|Ef\|_{L^6}^6$. On $\Phi^{-1}(0)$ one has $\Lambda = J(f)$, because

$$\Lambda \|f\|_2^2 = \operatorname{Re}\langle T(f),\ f\rangle = \int_{\mathbb{R}^2} |Ef(x)|^6\, dx.$$

If $t \mapsto (f(t), \Lambda(t))$ is a C^1 arc in $\Phi^{-1}(0)$, then $\|f(t)\|_2 = 1$ and $T(f(t)) = \Lambda(t) f(t)$, so

$$\frac{d}{dt} J(f(t)) = 6\operatorname{Re}\langle T(f(t)),\ f'(t)\rangle = 6\Lambda(t)\operatorname{Re}\langle f(t),\ f'(t)\rangle = 0.$$

Therefore $J(f(t))$, and hence $\Lambda(t)$, is constant on every such arc. Since the local analytic components are connected by finite chains of these arcs, Λ is constant on each local connected component of $\Phi^{-1}(0)$.

Finally, compactness of $\mathcal{C}_+$ gives a finite cover by neighbourhoods of the type described in Step 2. Each neighbourhood contains only finitely many local critical components, and Step 3 gives at most one value of Λ on each of them. Hence the projection of $\mathcal{C}_+$ onto the Λ coordinate, namely $\mathcal{V}_+(\Lambda_-, \Lambda_+)$, is finite. $\square$

4 Effective Spectral Gap Around the Constant Level

Let

$$f_0 = (2\pi)^{-1/2}, \qquad \Lambda_0 := \|Ef_0\|_{L^6}^6 = 16\pi^4 \int_0^\infty J_0(r)^6\, r\, dr.$$

If C_{ST} denotes any explicit Stein–Tomas bound for $E : L^2(S^1) \to L^6(\mathbb{R}^2)$, then every normalised critical value satisfies $\Lambda \le C_{\mathrm{ST}}^6$. Thus the natural positive window around the constant value is

$$W_0 := \left[\frac{1}{2}\Lambda_0,\ C_{\mathrm{ST}}^6\right].$$

By Theorem 3.14, the set $\mathcal{V}_+(W_0)$ of positive critical values in this window is finite and contains Λ_0. Hence

$$\gamma_0 := \operatorname{dist}\big(\Lambda_0,\, \mathcal{V}_+(W_0) \setminus \{\Lambda_0\}\big) \tag{15}$$

is strictly positive, with the convention $\gamma_0 = +\infty$ if $\mathcal{V}_+(W_0) = \{\Lambda_0\}$.

The qualitative proof gives no usable lower bound on γ_0. The central task is therefore to make the following implication effective:

$$T(f) = \Lambda f, \quad \|f\|_2 = 1, \quad |\Lambda - \Lambda_0| \ll 1 \quad \Longrightarrow \quad \operatorname{dist}(f, G \cdot f_0) \ll 1.$$

Once a solution is forced into a sufficiently small tube around the constant orbit, quantitative nondegeneracy at f_0 excludes it unless it is the constant itself.

4.1 Quantitative local isolation at the constant

On the slice orthogonal to phase and translations, define the linear nondegeneracy gap

$$\nu_0 := \min\left\{\frac{4}{5}\Lambda_0,\ \inf_{n\ge 2} |\mathcal{I}_n - \Lambda_0|\,,\ \inf_{n\ge 2} |5\mathcal{I}_n - \Lambda_0|\right\}, \qquad \mathcal{I}_n = 16\pi^4 \mathcal{J}_n.$$

The constant-solution nondegeneracy theorem gives $\nu_0 > 0$. For an effective radius one needs explicit constants:

$$\nu_0 \ge \underline{\nu} > 0, \qquad \left\|D\widetilde{F}(f) - D\widetilde{F}(g)\right\|_{X\to X} \le M_0\, \|f - g\|_2$$

on a gauge-fixed slice X near f_0, where $\widetilde{F}$ is the sliced Euler–Lagrange map. The standard Newton–Kantorovich estimate then gives the explicit uniqueness radius

$$r_0 = \frac{2\underline{\nu}}{M_0}. \tag{16}$$

Indeed, if $\widetilde{F}(f) = 0 = \widetilde{F}(f_0)$ and $\|f - f_0\|_2 < r_0$, then

$$\underline{\nu}\,\|f - f_0\|_2 \le \frac{M_0}{2}\,\|f - f_0\|_2^2\,,$$

which is impossible unless $f = f_0$.

Thus a quantitative spectral gap follows from any effective compactness statement which places all Λ-nearby solutions inside this ball. The remaining subsections describe two analytic ways to obtain such a statement.

4.2 Effective Fredholm-atlas route

Let

$$\mathcal{S}_\Lambda := \{f \ge 0 :\ \|f\|_2 = 1,\ T(f) = \Lambda f\}$$

in the positive gauge. The desired effective compactness statement is a modulus

$$\operatorname{dist}\big(\mathcal{S}_\Lambda, \mathcal{S}_{\Lambda_0}\big) \le \rho(|\Lambda - \Lambda_0|), \qquad \rho(t) \to 0 \quad (t \downarrow 0), \tag{17}$$

with explicit control of ρ. If, in addition,

$$\mathcal{S}_{\Lambda_0}/G = \{[f_0]\},$$

then choosing $\eta > 0$ so that $\rho(\eta) < r_0$ proves $\gamma_0 \ge \eta$.

One way to make (17) quantitative is to revisit the proof of finite eigenvalue count with constants. Fix a small window

$$I_\eta = [\Lambda_0 - \eta, \Lambda_0 + \eta].$$

The ingredients needed are the following.

1. **Effective analytic a priori bounds.** For every solution with $\Lambda \in I_\eta$, prove explicit estimates in a fixed analytic Banach space, for example

   $$\|f\|_{\mathcal{A}_\rho} \le A(\eta),$$

   where $\mathcal{A}_\rho$ denotes functions holomorphic in a complex strip $\{|\operatorname{Im}\theta| < \rho\}$ with a Banach norm such as the sup norm on the strip. The point is not to truncate Fourier series, but to put the whole solution set in one compact analytic class. The inclusion $\mathcal{A}_\rho \hookrightarrow \mathcal{A}_{\rho'}$ is compact for $\rho' < \rho$, and the constants $A(\eta), \rho, \rho'$ give a computable compactness modulus for the possible shapes.

2. **Effective finite Fredholm atlas at level Λ_0.** Cover $\mathcal{S}_{\Lambda_0}$ by finitely many gauge-fixed charts. At a chart centre f_*, write the constrained equation as

$$F(f,\Lambda)=0, \qquad F(f,\Lambda):=T(f)-\Lambda f, \qquad \|f\|_2=1,$$

on a slice. Since $D_fF(f_*,\Lambda_0)$ is Fredholm, split the slice as

$$X=K_*\oplus X_*, \qquad K_*:=\ker D_fF(f_*,\Lambda_0),$$

and solve the X_* component by a quantitative inverse function theorem. This reduces the local zero set to a finite-dimensional analytic equation

$$\Phi_*(a,\Lambda)=0, \qquad a\in K_*.$$

The number of active parameters is therefore bounded by $\dim K_*$, which is finite. An effective proof needs an explicit upper bound for these kernel dimensions, or a direct computable basis for the kernels in the relevant atlas.

3. **Quantitative inverse bounds off the singular set.** Away from the singular locus

$$\Sigma_{\Lambda_0}:=\{f\in\mathcal{S}_{\Lambda_0}: D_fF(f,\Lambda_0) \text{ is not invertible on the slice}\},$$

one needs a lower bound

$$\|D_fF(f,\Lambda_0)h\|_2\geq m(\delta)\,\|h\|_2 \tag{18}$$

whenever $\operatorname{dist}(f,\Sigma_{\Lambda_0})\geq\delta$. Together with a Lipschitz bound for D_fF, this gives a quantitative local inverse theorem: solution sets at nearby Λ move by at most $O(|\Lambda-\Lambda_0|/m(\delta))$ in these regular charts.

4. **Effective treatment of singular charts.** Near Σ_{Λ_0}, the Lyapunov–Schmidt maps $\Phi_*(a,\Lambda)$ are finite-dimensional real-analytic systems. A quantitative Lojasiewicz or Weierstrass-preparation estimate would bound how branches at $\Lambda\neq\Lambda_0$ can leave the singular set. If the only singular directions are symmetry directions and the slice has removed them, this step collapses to the ordinary inverse theorem.

This is the precise form of the "skewering" strategy. The a priori bounds make the window compact in an analytic topology with a computable modulus; the Fredholm atlas reduces the infinite-dimensional problem to finitely many local finite-dimensional systems; inverse estimates away from the singular set control the motion of solution sets as Λ varies; and the singular charts are handled by finite-dimensional analytic estimates. The output is an explicit function ρ in (17). Combining (17) with the local radius (16) gives an effective nonlinear spectral gap.

A stronger but cleaner sufficient input is the following stability inequality:

$$\Lambda_0-J(f)\geq c_*\operatorname{dist}(f,G\cdot f_0)^2, \qquad J(f):=\|Ef\|_6^6, \tag{19}$$

for all nonnegative normalised f in the relevant window. This is a Bianchi–Egnell type statement for the sharp restriction inequality. If (19) is available, the Fredholm-atlas machinery is unnecessary for the sharp constant: any nonlinear eigenpair with Λ close to Λ_0 is forced directly into the local nondegeneracy ball.

4.3 What has to be made effective in one chart

We spell out the local reduction because this is the point at which the infinite-dimensional problem becomes a finite-dimensional question about zero sets. Work in a gauge-fixed analytic Banach space X for the unknown and an analytic target space Y for the residual. At a solution (f_*, Λ_0) set

$$F(f, \Lambda) := T(f) - \Lambda f, \qquad L_* := D_f F(f_*, \Lambda_0).$$

After imposing the normalisation and gauge conditions, $L_* : X \to Y$ is Fredholm. Choose splittings

$$X = K_* \oplus X_*, \qquad K_* = \ker L_*, \qquad Y = Y_* \oplus C_*,$$

with C_* finite-dimensional and $L_* : X_* \to Y_*$ an isomorphism. Let $Q_* : Y \to Y_*$ and $\Pi_* : Y \to C_*$ be the corresponding projections. Writing

$$f = f_* + a + w, \qquad a \in K_*, \quad w \in X_*,$$

the equation splits into

$$Q_* F(f_* + a + w, \Lambda) = 0, \qquad \Pi_* F(f_* + a + w, \Lambda) = 0.$$

If

$$B_* := \left\| (Q_* L_*|_{X_*})^{-1} \right\|_{Y_* \to X_*}$$

is explicit and the derivative of $Q_* F$ is Lipschitz with constant M_* on a ball of radius r_*, then the quantitative inverse function theorem solves the first equation as

$$w = \Psi_*(a, \Lambda)$$

for $\|a\|_{K_*} + |\Lambda - \Lambda_0| \le c/(B_* M_*)$. The remaining equation is finite-dimensional:

$$\Phi_*(a, \Lambda) := \Pi_* F(f_* + a + \Psi_*(a, \Lambda), \Lambda) = 0, \qquad a \in K_*. \tag{20}$$

Thus the constants needed from the Fredholm reduction are concrete:

$$\dim K_*, \qquad B_*, \qquad M_*, \qquad r_*, \qquad \|\partial_\Lambda \Phi_*\| .$$

Away from singular points of $\Phi_*(\cdot, \Lambda_0)$, a lower bound on the smallest singular value of $D_a \Phi_*$ gives the local motion of zero sets by the ordinary inverse theorem. At singular points, the required replacement is an effective finite-dimensional analytic estimate, such as a Lojasiewicz distance inequality or a Weierstrass-preparation bound for Φ_*. This is now a problem about finitely many analytic maps on compact subsets of Euclidean space, not a problem about global Fourier tails.

4.4 Quantifying the finite-dimensional zero-set constants

We now isolate the finite-dimensional problem produced by the previous subsection. Fix one Lyapunov–Schmidt chart and write

$$\Phi(a,\lambda)=0, \qquad a\in U\subset\mathbb{R}^d, \qquad \lambda\in I_\eta.$$

Let

$$Z_0:=\{a\in K:\Phi(a,\Lambda_0)=0\}, \qquad K\Subset U$$

be the part of the level-Λ_0 zero set covered by the chart. To control nearby levels, it is enough to prove an effective distance inequality

$$\operatorname{dist}(a,Z_0)\le C_{\mathrm{Loj}}\,\|\Phi(a,\Lambda_0)\|^{\theta} \qquad (a\in K), \tag{21}$$

together with a bound

$$\|\Phi(a,\lambda)-\Phi(a,\Lambda_0)\|\le L_\lambda|\lambda-\Lambda_0| \qquad (a\in K,\ \lambda\in I_\eta). \tag{22}$$

Then every zero $a_\lambda\in K$ of $\Phi(\cdot,\lambda)$ satisfies

$$\operatorname{dist}(a_\lambda,Z_0)\le C_{\mathrm{Loj}}L_\lambda^{\theta}|\lambda-\Lambda_0|^{\theta}. \tag{23}$$

This is the local quantitative upper-semicontinuity estimate. The lift from a-space back to the function space only costs the Lipschitz constant of the chart map

$$a,\lambda\mapsto f_*+a+\Psi(a,\lambda).$$

Proposition 4.1 (One-chart spectral-gap modulus)**.** *Suppose a chart satisfies* (21) *and* (22)*. Suppose also that the chart map satisfies*

$$\|\Psi(a,\lambda)-\Psi(b,\Lambda_0)\|_X\le L_a\,\|a-b\|+L'_\lambda|\lambda-\Lambda_0|$$

on the chart. If the level-Λ_0 zero set in the chart is contained in the constant orbit and the local exclusion radius around that orbit is r_0, then this chart contains no nonconstant solution with $0<|\lambda-\Lambda_0|<\eta_*$, where any $\eta_*>0$ satisfying

$$L_aC_{\mathrm{Loj}}L_\lambda^{\theta}\eta_*^{\theta}+L'_\lambda\eta_*<r_0 \tag{24}$$

is admissible.

Proof. Let a_λ be a zero of $\Phi(\cdot,\lambda)$. By (23), choose $a_0\in Z_0$ with

$$\|a_\lambda-a_0\|\le C_{\mathrm{Loj}}L_\lambda^{\theta}|\lambda-\Lambda_0|^{\theta}.$$

The corresponding functions satisfy

$$\|f(a_\lambda,\lambda)-f(a_0,\Lambda_0)\|_X\le L_aC_{\mathrm{Loj}}L_\lambda^{\theta}|\lambda-\Lambda_0|^{\theta}+L'_\lambda|\lambda-\Lambda_0|.$$

If $|\lambda-\Lambda_0|<\eta_*$, condition (24) places the solution inside the local exclusion ball around the level-Λ_0 zero set. Local nondegeneracy then forces the solution to be the constant one, hence $\lambda=\Lambda_0$. □

Thus the problem is reduced to obtaining explicit C_{Loj} and θ for each finite-dimensional analytic map $\Phi(\cdot,\Lambda_0)$. There are two regimes.

Regular charts. If

$$s_* := \inf_{a \in K \cap Z_0} \sigma_{\min}\big(D_a\Phi(a,\Lambda_0)|_{N_a Z_0}\big) > 0,$$

where $N_a Z_0$ is a chosen normal complement to the smooth zero set, then $\theta = 1$. If $D_a\Phi$ is M_Φ-Lipschitz on a tubular neighbourhood of radius r around Z_0, the quantitative inverse theorem gives

$$\operatorname{dist}(a, Z_0) \le 2s_*^{-1} \|\Phi(a,\Lambda_0)\|$$

whenever $\operatorname{dist}(a, Z_0) \le s_*/M_\Phi$. Outside this tube, compactness gives a positive residual floor

$$m_{\text{far}} := \inf_{\substack{a \in K \\ \operatorname{dist}(a,Z_0) \ge s_*/M_\Phi}} \|\Phi(a,\Lambda_0)\|.$$

Combining the two regions gives an explicit global constant on K:

$$C_{\text{Loj}} = \max\left\{2s_*^{-1},\ \frac{\operatorname{diam} K}{m_{\text{far}}}\right\}, \qquad \theta = 1.$$

All quantities here are finite-dimensional and come from the Lyapunov–Schmidt map itself.

Singular charts. At a singular zero $a_* \in Z_0$, the inverse theorem no longer gives $\theta = 1$. The effective replacement is a finite-jet non-flatness estimate. One needs a radius $r_* > 0$, an integer $q \ge 1$, and a constant $\kappa_* > 0$ such that on the ball $B(a_*, r_*)$ the ideal generated by the components of $\Phi_0(a) := \Phi(a,\Lambda_0)$ has nontrivial normal order at most q, quantitatively bounded below by κ_*. Concretely, after choosing analytic coordinates $a = (s, n)$, where s is tangent to the local zero set and n is normal, this means that some finite collection of derivatives

$$\partial_n^\alpha \Phi_{0,j}(s, 0), \qquad 1 \le |\alpha| \le q,$$

generates the normal ideal and has a controlled lower bound κ_* on the relevant s-patch.

Together with a Cauchy bound

$$\sup_{B_{\mathbb{C}}(a_*, 2r_*)} \|\Phi_0\| \le H_*$$

on the complexified polydisc, quantitative Weierstrass preparation or quantitative Lojasiewicz gives a bound of the form

$$\operatorname{dist}(a, Z_0) \le C(d, q, r_*, H_*, \kappa_*)\ \|\Phi_0(a)\|^{1/q'} \tag{25}$$

on $B(a_*, r_*/2)$, where q' is controlled by d and q. The exact exponent is less important than effectivity: once C and q' are explicit, Proposition 4.1 turns them into a gap modulus. The true obstruction is therefore not Fredholmness, but proving a quantitative finite-jet non-flatness bound for the reduced analytic map at the singular charts.

What must be proved for the present problem. For the restriction problem near Λ_0, the cleanest outcome is: after quotienting symmetries and imposing positivity, every level-Λ_0 chart is regular and

$$Z_0/G = \{[f_0]\}.$$

Then every chart has $\theta = 1$, and the spectral gap is controlled by ordinary inverse-function constants. If singular charts remain, one must compute or estimate:

$$d = \dim K_*, \qquad q, \qquad r_*, \qquad H_*, \qquad \kappa_*, \qquad L_\lambda, \qquad L_a,\ L'_\lambda.$$

These constants are produced entirely by the Fredholm reduction: d and B_* come from the kernel and the inverse on the complement; H_*, M_*, L_λ, L_a, and L'_λ come from Cauchy estimates for the analytic map F and the quantitative inverse function theorem; and q, κ_* are finite-jet data of the reduced map Φ_*. This is the finite-dimensional chart-constant problem which replaces any tail-based strategy.

Dictionary of constants and expected difficulty. We record the precise role of the constants, since the effectiveness of the spectral gap depends on which of them can be obtained directly from the Fredholm reduction and which require genuinely new input.

The constant

$$B_* = \left\| \left(D_f F(f_*, \Lambda_0)|_{X_*} \right)^{-1} \right\|$$

is the inverse bound on the Fredholm complement after the splittings

$$X = K_* \oplus X_*, \qquad Y = C_* \oplus Y_*.$$

It measures the cost of solving the infinite-dimensional equation orthogonally to the kernel. The integer

$$d = \dim K_*$$

is the number of variables in the reduced equation $\Phi(a, \lambda) = 0$. These two constants are the most intrinsic Fredholm constants. At the constant solution they should be among the most accessible ones, because the linearized operator is expected to decompose into the natural symmetry and angular modes.

The constants $H_*, r_*, M_\Phi, L_\lambda, L_a, L'_\lambda$ are analytic chart constants. A typical choice of H_* is a Cauchy bound on a complexified chart,

$$\sup_{B_{\mathbb{C}}(a_*, 2r_*)} \|\Phi(a, \Lambda_0)\| \le H_*.$$

The radius r_* is the radius on which the implicit map $\Psi(a, \lambda)$ and the reduced map $\Phi(a, \lambda)$ are defined with the desired estimates. The constant M_Φ controls the variation of the derivative of the reduced map,

$$\|D_a\Phi(a, \Lambda_0) - D_a\Phi(b, \Lambda_0)\| \le M_\Phi \|a - b\|.$$

The parameter Lipschitz constant L_λ is defined by

$$\|\Phi(a,\lambda) - \Phi(a,\Lambda_0)\| \le L_\lambda |\lambda - \Lambda_0|,$$

whereas L_a and L'_λ control the lift from reduced coordinates to the function space,

$$\|f(a,\Lambda_0) - f(b,\Lambda_0)\|_X \le L_a \|a-b\|, \qquad \|f(a,\lambda) - f(a,\Lambda_0)\|_X \le L'_\lambda |\lambda - \Lambda_0|.$$

These constants should be obtainable, at least in principle, from Banach-algebra estimates for the analytic norm, convolution bounds for the fixed kernel, Cauchy estimates, and the quantitative implicit function theorem. They may be numerically unattractive, but they do not encode the main geometric obstruction.

The next constants express regularity of the level-Λ_0 zero set. The number

$$s_* = \inf_{a \in K \cap Z_0} \sigma_{\min}\big(D_a\Phi(a,\Lambda_0)|_{N_a Z_0}\big)$$

is the smallest normal singular value of the reduced equation along Z_0. If $s_* > 0$, the chart is regular in the normal directions and one has the Lojasiewicz exponent $\theta = 1$. The radius r_0 is the local exclusion radius around the constant orbit: within this radius, any level-Λ_0 solution must be the constant solution, modulo the imposed symmetries. Both s_* and r_0 should be accessible if one can prove that the level-Λ_0 solution set is locally just the constant orbit after quotienting symmetries and imposing positivity. They are harder than the pure chart constants, but they are still ordinary nondegeneracy constants rather than singular analytic-geometry constants.

The residual floor

$$m_{\text{far}} = \inf_{\substack{a \in K \\ \operatorname{dist}(a,Z_0) \ge s_*/M_\Phi}} \|\Phi(a,\Lambda_0)\|$$

patches the local inverse estimate near Z_0 into a global estimate on the compact chart. Although this is finite-dimensional, it is not a cheap constant: a quantitative lower bound for m_{far} is already a finite-dimensional isolation statement away from the known zero set.

Finally, the genuinely difficult constants are the singular Lojasiewicz data. The inequality

$$\operatorname{dist}(a,Z_0) \le C_{\text{Loj}} \|\Phi(a,\Lambda_0)\|^\theta$$

is immediate from compact real-analytic geometry, but making C_{Loj} and θ explicit is easy only in regular charts. In singular charts one needs a finite order q and a non-flatness lower bound κ_*, schematically

$$\max_{1 \le |\alpha| \le q, j} |\partial_n^\alpha \Phi_{0,j}(s,0)| \ge \kappa_*,$$

on the relevant tangential patch. Together with H_* and r_*, these data feed quantitative Weierstrass preparation or quantitative Lojasiewicz estimates and produce an exponent

of the form $\theta = 1/q'$. This is the hardest part of the program if singular charts cannot be ruled out.

Thus the practical hierarchy is as follows. The constants

$$d,\ B_*,\ H_*,\ r_*,\ M_\Phi,\ L_\lambda,\ L_a,\ L'_\lambda$$

are bookkeeping constants of the Fredholm reduction and analytic implicit function theorem. The constants

$$s_*,\ r_0,\ m_{\text{far}}$$

are quantitative isolation and nondegeneracy constants for the level-Λ_0 solution set. The constants

$$q,\ \kappa_*,\ C_{\text{Loj}},\ \theta$$

are the singular analytic-geometry constants. The preferred route is therefore to prove that, after quotienting symmetries and imposing positivity, the relevant level-Λ_0 charts are regular. Then $\theta = 1$, C_{Loj} is controlled by ordinary inverse function constants, and the gap follows from

$$L_a C_{\text{Loj}} L_\lambda^\theta \eta_*^\theta + L'_\lambda \eta_* < r_0.$$

If singular charts remain, the program is still finite-dimensional, but one must supply effective finite-jet non-flatness estimates for $\Phi(\cdot, \Lambda_0)$.

4.5 Residual functional and distance to zero sets

Another way to package the preceding plan is to study a residual functional. Let X be a normed target space in which the residual is defined and in which small residual has geometric meaning. Formally, one would like to use

$$\mathcal{E}_X(f, \lambda) := \|T(f) - \lambda f\|_X^2, \qquad \lambda \in \mathbb{R}, \qquad \|f\|_2 = 1. \tag{26}$$

Zeros of $\mathcal{E}_X$ are precisely nonlinear eigenpairs. If X is chosen so that $T(f) - \lambda f$ is analytic in f and λ, then the zero set is an analytic variety, at least after restricting to the analytic regularity class supplied by the bootstrap.

There is a small functional-analytic caveat. If f is only assumed to lie in L^2, then a strong analytic norm X need not contain the term λf. Thus one should either take $X = L^2$ or a negative Sobolev space for the global residual, or first use the a priori bootstrap to place all relevant solutions in a fixed analytic Banach space $\mathcal{A}_\rho$ and then define $\mathcal{E}_X$ there. In the present route, finite-dimensionality enters only locally, through Fredholm Lyapunov–Schmidt reduction, not through a global Fourier cut-off.

The essential quantitative question is then:

how can one bound the distance to $Z(P) := \{P = 0\}$ directly from $\|P\|$?

The standard answer is a quantitative Lojasiewicz inequality. In a Fredholm chart, the infinite-dimensional equation is reduced to

$$\Phi(a, \lambda) = 0, \qquad a \in U \subset \mathbb{R}^d,$$

where Φ is real-analytic. On a compact set $K \Subset U$, one seeks explicit constants $C, \theta > 0$ such that

$$\operatorname{dist}(a, Z(\Phi_{\Lambda_0}) \cap K) \leq C \left\|\Phi(a, \Lambda_0)\right\|^{\theta}, \qquad a \in K, \tag{27}$$

where $\Phi_{\Lambda_0}(a) := \Phi(a, \Lambda_0)$. If $a \in Z(\Phi_\lambda)$, then

$$\Phi(a, \Lambda_0) = \Phi(a, \Lambda_0) - \Phi(a, \lambda),$$

so an explicit bound on $\partial_\lambda \Phi$ gives

$$\|\Phi(a, \Lambda_0)\| \leq L_\lambda |\lambda - \Lambda_0|.$$

Substituting in (27) yields

$$\operatorname{dist}\big(a, Z(\Phi_{\Lambda_0}) \cap K\big) \leq C L_\lambda^{\theta} |\lambda - \Lambda_0|^{\theta}. \tag{28}$$

This is exactly the local form of the effective upper-semicontinuity modulus (17).

Thus the residual-functional route reduces the nonlinear spectral gap to three explicit analytic estimates:

1. a bootstrap placing all relevant eigenfunctions in a compact analytic class, together with finitely many gauge-fixed Fredholm charts;
2. in each Lyapunov–Schmidt chart, an effective Lojasiewicz or Weierstrass-preparation bound for the analytic residual $\Phi(\cdot, \Lambda_0)$ on the relevant compact set;
3. effective local rigidity of the zero set at level Λ_0, ideally $\mathcal{S}_{\Lambda_0}/G = \{[f_0]\}$ in the positive gauge.

If these are available, then the spectral gap is quantified by choosing $|\lambda - \Lambda_0|$ so small that the Lojasiewicz distance to $\mathcal{S}_{\Lambda_0}$ is below the local isolation radius r_0.

4.6 A projected-equation route to the gap

We now record a more analytic route which avoids a blind numerical search over the annulus. The guiding principle is that a critical point with Λ close to Λ_0 should be forced into the local nondegeneracy ball around f_0. Once this is known, the Newton–Kantorovich/local uniqueness argument excludes it unless it is exactly f_0.

Throughout this subsection we work in the real Hilbert space underlying $L^2(S^1)$, with inner product $\operatorname{Re}\langle \cdot, \cdot \rangle$, and write

$$T(f) := E^*(|Ef|^4 Ef).$$

Thus critical points satisfy

$$T(f) = \Lambda f, \qquad \|f\|_2 = 1, \qquad \Lambda = J(f) := \|Ef\|_6^6 .$$

Gauge and coercive slice. Let $f_0 = (2\pi)^{-1/2}$. The tangent space to the symmetry orbit at f_0 is generated by the phase direction if_0 and the two translation directions, which are the $n = \pm 1$ Fourier modes. For the positive problem the phase is fixed, and we assume that the translation gauge has been chosen so that the perturbation is orthogonal to the translation directions. In this gauge write

$$f = af_0 + \xi, \qquad a = \operatorname{Re}\langle f,\, f_0\rangle > 0, \qquad \xi \perp f_0, \quad \xi \perp T_{f_0}(G \cdot f_0). \tag{29}$$

Since $\|f\|_2 = 1$, we have

$$a^2 + \sigma^2 = 1, \qquad \sigma := \|\xi\|_2 .$$

The point of the gauge is that ξ belongs to the spectral subspace on which the linearised operator has a genuine gap.

Linear spectral input. Let $K := DT(f_0)$. On the coercive slice define

$$\mu_0 := \sup_{\substack{\xi \perp f_0,\ \xi \perp T_{f_0}(G\cdot f_0) \\ \|\xi\|_2 = 1}} \operatorname{Re}\langle K\xi,\, \xi\rangle .$$

By the Fourier decomposition in Proposition 2.8,

$$\mu_0 = \max_{n \geq 2} 5\mathcal{I}_n.$$

The certified Bessel gap gives, for example,

$$\mu_0 \leq 0.6\Lambda_0.$$

For the sharp local constant one expects

$$\mu_0 = 5\mathcal{I}_2, \qquad \Lambda_0 - \mu_0 \approx 0.452\Lambda_0.$$

Only the upper bound $\mu_0 < \Lambda_0$ is needed for the abstract argument below.

Finite Taylor expansion of T. Since T is a homogeneous polynomial map of degree five on the real Hilbert space, the Taylor expansion around f_0 terminates:

$$T(f_0 + h) = \Lambda_0 f_0 + Kh + \sum_{j=2}^{5} \frac{1}{j!} D^j T(f_0)[h^j]. \tag{30}$$

For $f = af_0 + \xi$ it is slightly cleaner to factor out the exact homogeneity:

$$T(af_0 + \xi) = a^5 T\left(f_0 + \frac{\xi}{a}\right) \qquad (a > 0).$$

Using (30), this gives

$$T(af_0 + \xi) = a^5 \Lambda_0 f_0 + a^4 K\xi + \sum_{j=2}^{5} a^{5-j} \frac{1}{j!} D^j T(f_0)[\xi^j]. \tag{31}$$

Remainder constants. For $2 \le j \le 5$, define the coercive-slice multilinear constants

$$C_j := \sup_{\substack{\xi \perp f_0,\ \xi \perp T_{f_0}(G\cdot f_0) \\ \|\xi\|_2 = 1}} \left| \operatorname{Re} \left\langle \frac{1}{j!} D^j T(f_0)[\xi^j],\, \xi \right\rangle \right|. \tag{32}$$

More generally, one may replace C_j by the operator norm of the $(j+1)$-linear form

$$(\xi_1, \dots, \xi_j, \zeta) \mapsto \operatorname{Re} \left\langle \frac{1}{j!} D^j T(f_0)[\xi_1, \dots, \xi_j],\, \zeta \right\rangle$$

on the coercive slice. The diagonal constants in (32) are the smallest constants needed for the following scalar proof.

Each C_j is in principle an explicit finite/infinite sum of Bessel product integrals. For example, C_2 involves integrals of the form

$$\int_0^\infty J_0(r)^3 J_{n_1}(r) J_{n_2}(r) J_{n_3}(r)\, r\, dr,$$

with angular selection rules. The higher C_j's involve the same sixfold Bessel product family appearing in the linear and multilinear expansions at the constant. The analytic proof only needs upper bounds for these constants; any computer assistance would enter only as certified evaluation of explicit Bessel-product norms.

The projected equation. Assume now that $f = af_0 + \xi$ is a positive normalised critical point in the above gauge:

$$T(f) = \Lambda f.$$

Take the real inner product with ξ. Since $\xi \perp f_0$, the constant term in (31) drops out and we obtain

$$\begin{aligned} \Lambda\sigma^2 &= a^4 \operatorname{Re} \langle K\xi,\, \xi \rangle + \sum_{j=2}^{5} a^{5-j} \operatorname{Re} \left\langle \frac{1}{j!} D^j T(f_0)[\xi^j],\, \xi \right\rangle \\ &\le a^4 \mu_0 \sigma^2 + C_2 a^3 \sigma^3 + C_3 a^2 \sigma^4 + C_4 a \sigma^5 + C_5 \sigma^6. \end{aligned} \tag{33}$$

This is the central inequality. It is purely analytic and contains only the linear spectral constant μ_0 and the explicit multilinear constants $C_2, \dots, C_5$.

Proposition 4.2 (Projected-equation exclusion criterion)**.** *Let $0 < r_0 < 1$ be a local uniqueness radius around f_0 in the positive gauge, and let $\eta > 0$. Suppose that for every $\sigma \in [r_0, 1]$, with $a = \sqrt{1-\sigma^2}$, one has*

$$(\Lambda_0 - \eta - a^4 \mu_0)\sigma^2 > C_2 a^3 \sigma^3 + C_3 a^2 \sigma^4 + C_4 a \sigma^5 + C_5 \sigma^6. \tag{34}$$

Then there is no positive normalised critical point $f \neq f_0$, modulo symmetries, with

$$|\Lambda - \Lambda_0| \leq \eta.$$

In particular, the nonlinear critical-value gap around Λ_0 satisfies

$$\gamma_0 \geq \eta.$$

Proof. Suppose $T(f) = \Lambda f$, $\|f\|_2 = 1$, $f \geq 0$, and $|\Lambda - \Lambda_0| \leq \eta$. Put f in the positive translation gauge and decompose it as in (29). If $\sigma < r_0$, local uniqueness gives $f = f_0$, hence $\Lambda = \Lambda_0$. Otherwise $\sigma \in [r_0, 1]$.

Since $\Lambda \geq \Lambda_0 - \eta$, the left side of (33) is bounded below by $(\Lambda_0 - \eta)\sigma^2$. Hence

$$(\Lambda_0 - \eta)\sigma^2 \leq a^4\mu_0\sigma^2 + C_2a^3\sigma^3 + C_3a^2\sigma^4 + C_4a\sigma^5 + C_5\sigma^6,$$

contradicting (34). Thus no such f exists outside the local ball, and the local ball contains only f_0. □

Remark 4.3 (Why this proves exact-level uniqueness). Taking $\eta = 0$ in (34) gives a direct uniqueness criterion at the exact value Λ_0:

$$T(f) = \Lambda_0 f, \quad \|f\|_2 = 1, \quad f \geq 0 \quad \Longrightarrow \quad f = f_0$$

modulo symmetries. Taking $\eta > 0$ gives the stronger quantitative gap statement.

Remark 4.4 (A usable sufficient form). Since $a \leq 1$, a cruder but simpler sufficient condition is

$$\Lambda_0 - \eta - \mu_0 > C_2\sigma + C_3\sigma^2 + C_4\sigma^3 + C_5\sigma^4 \qquad (r_0 \leq \sigma \leq 1). \tag{35}$$

This form is usually too wasteful for large σ, but it shows the mechanism clearly: the linear spectral gap $\Lambda_0 - \mu_0$ must dominate the nonlinear remainders on the complement of the local ball. A sharper implementation should use (34), because the factor $a = \sqrt{1 - \sigma^2}$ suppresses the lower-order nonlinear terms away from the constant mode.

What remains to make the criterion effective. The proof of Proposition 4.2 reduces the nonlinear gap problem to the following finite list of quantitative inputs:

1. a certified local uniqueness radius r_0, obtained from ν_0 and the Newton–Kantorovich Lipschitz constant;

2. the linear coercive constant

$$\mu_0 = \sup_{\xi \in \mathcal{S},\ \|\xi\|=1} \operatorname{Re} \langle K\xi,\, \xi \rangle,$$

   where $\mathcal{S}$ is the slice orthogonal to f_0 and the symmetry directions;

3. certified upper bounds for the multilinear constants $C_2, \ldots, C_5$;

4. a one-variable verification of (34) on $\sigma \in [r_0, 1]$.

This is the desired "straight proof" architecture: the argument itself is analytic, while numerical work is confined to explicit Bessel-product constants and a final one-dimensional inequality.

5 Options for Completing the Sharp Constant

Two paths are available, depending on the desired strength of the result.

Path 1: Identify $\mathcal{R}$ (the sharp constant).

1. **Verify Bessel integrals.** Check $\mathcal{J}_n \neq \mathcal{J}_0$ and $5\mathcal{J}_n \neq \mathcal{J}_0$ for $n = 2, 3, \ldots, n_*$ by interval arithmetic. This establishes nondegeneracy at the constant solution. *Routine numerical computation.*

2. **Restrict the search to nonnegative eigenfunctions.** By Lemma 2.1, any extremiser can be replaced by a nonnegative extremiser, possibly in a different shape. Therefore, for the sharp constant, one only needs to retain those candidate values Λ_j for which the nonlinear eigenvalue equation has at least one nonnegative solution. The positive gauge also removes exact translation ambiguity by Lemma 3.11, and Theorem 3.14 proves that the retained critical values in any fixed positive window are finite.

3. **Show the constant is the unique maximiser.** Theorem 3.7 gives existence of a maximiser. If one can prove that the constant function $f_0 = (2\pi)^{-1/2}$ is the unique maximiser modulo G—e.g., by a rearrangement or symmetrisation argument, or by ruling out non-radial maximisers via the eigenvalue analysis—then $\mathcal{R} = \|Ef_0\|_{L^6}$ and the programme is complete. *This bypasses compactness of the full critical set.*

Path 2: Finiteness of all critical values above ε (the full structural result).

1. Bessel integrals (same as above).

2. **Rule out multi-profile splitting (Theorem 3.10).** The profile decomposition is in place (Theorem 3.3), and each profile inherits the Euler–Lagrange equation (Proposition 3.6). The open step is to show that no critical sequence admits more than one profile. Section 3.1 identifies three general strategies: descending induction, Pohozaev–Morawetz monotonicity, and unique continuation for cross terms. *Requires new analytic input.*

3. **Enumerate non-constant shapes (if any).** Either prove that the constant is the only critical shape, or verify nondegeneracy at all other shapes.